

NS6

A sphere of radius a is immersed in a liquid of density ρ , viscosity μ , thermal conductivity k , and specific heat c . The liquid is sucked with radial velocity $v_r = -V_0$ through the porous surface of the sphere. For the buoyancy-free, steady sphericallysymmetrical motion induced in the surrounding liquid,

1. Obtain the velocity $v_r(r)$
2. Compute the trajectory of a fluid particle initially located at $r = r_i > a$, as well as the time t_i at which the fluid particle is sucked by the sphere.
3. Determine the pressure distribution $p - p_\infty$, where p_∞ is the pressure value far from the sphere.
4. Calculate the temperature distribution in the liquid $T(r)$ assuming that the temperature of the sphere is T_s , smaller then the value T_∞ found in the surrounding ambient atmosphere.
5. Obtain the amount of heat transfered to the sphere per unit time.



1) $\text{con: } \frac{1}{r^2} \frac{d}{dr}(r^2 v_r) = 0 \Rightarrow r^2 v_r = C = -a^2 V_0 \Rightarrow \boxed{v_r = -\frac{a^2}{r^2} V_0}$

2) $\frac{dr}{dt} = v_r = -\frac{a^2}{r^2} V_0 \Rightarrow \boxed{\frac{r^3 - r_i^3}{3} = -a^2 V_0 t} \quad r=a, \quad \boxed{t=t_i = \frac{r_i^3 - a^3}{3a^2 V_0}}$

3) r-mom: $\rho v_r \frac{dv_r}{dr} = -\frac{dp}{dr} \Rightarrow P + \rho \frac{v_r^2}{2} = C \rightarrow \boxed{P + \rho \frac{a^4}{r^4} \frac{V_0^2}{2} = P_\infty}$

4) $\rho c v_r \frac{dT}{dr} = \phi_r + \frac{k}{r^2} \frac{d}{dr}(r^2 \frac{dT}{dr})$, $\phi_r = 2\mu \left[\left(\frac{dv_r}{dr} \right)^2 + 2 \left(\frac{v_r}{r} \right)^2 \right] = 12\mu \frac{V_0^2 a^4}{r^6}$

$\frac{d}{dr}(r^2 \frac{dT}{dr}) + \frac{\rho c V_0}{k} \frac{a^2}{r} \frac{dT}{dr} + \frac{12\mu}{k} \frac{V_0^2 a^2}{r^2} = 0 \Rightarrow T = C_1 e^{\frac{\rho c V_0 a}{k} \frac{a}{r}} + C_2 + \frac{12\mu V_0}{\rho c r}$

$$T = T_\infty + \frac{12\mu V_0}{\rho c r} + (T_s - T_\infty - \frac{12\mu V_0}{\rho c a}) \frac{(e^{\frac{\rho c V_0 a}{k} \frac{a}{r}} - 1)}{(e^{\frac{\rho c V_0 a}{k}} - 1)}$$



$$\left(\vec{q} \cdot \vec{n} = k \frac{\partial T}{\partial r} \right)_{r=a} = \left[-\frac{\rho c V_0 (T_s - T_\infty - \frac{12\mu V_0}{\rho c a})}{e^{\frac{\rho c V_0 a}{k}} - 1} - k \frac{12\mu V_0}{\rho c a^2} \right] 4\pi a^2$$